

- 1(a) As part of a research project, the masses, m grams, of a random sample of 1000 pebbles from a certain beach were recorded. The results are summarised in the table.

Mass (g)	$50 \leq m < 150$	$150 \leq m < 200$	$200 \leq m < 250$	$250 \leq m < 350$
Frequency	162	318	355	165

Calculate estimates of the mean and standard deviation of these masses.

[2]

- (b) The masses, x grams, of a random sample of 1000 pebbles on a different beach were also found. It was proposed that the distribution of these masses should be modelled by the random variable $X \sim N(200, 3600)$.

Use the model to find $P(150 < X < 210)$.

[1]

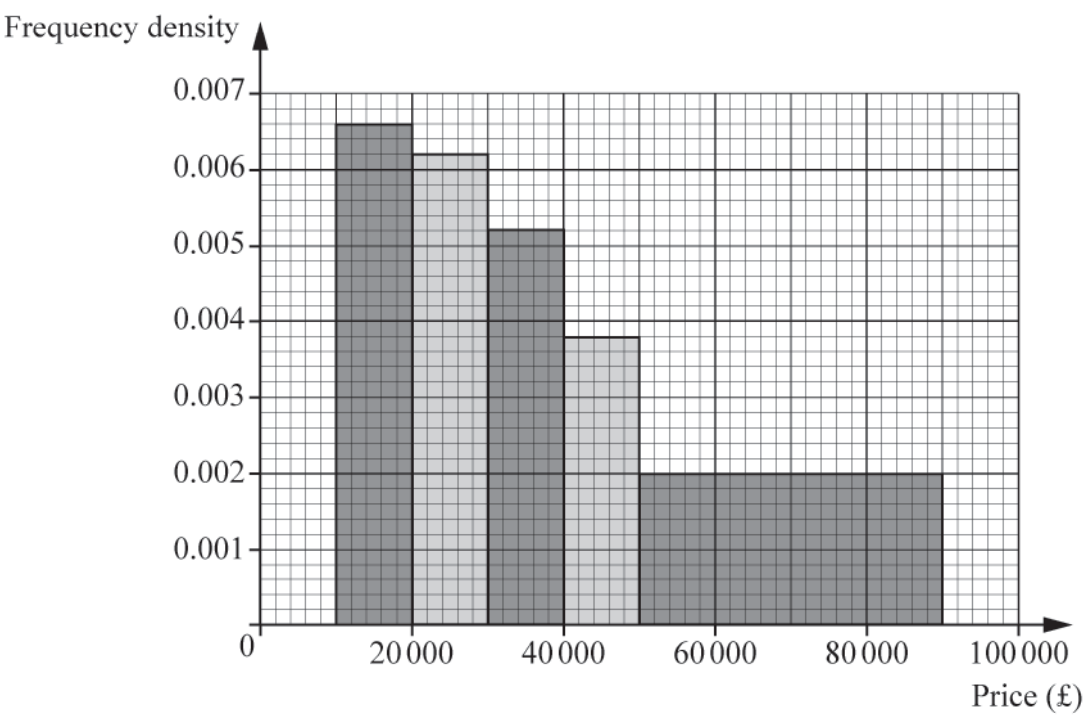
- (c) Use the model to determine x_1 such that $P(160 < X < x_1) = 0.6$, giving your answer correct to **five** significant figures. [3]

- (d) It was found that the smallest and largest masses of the pebbles in this second sample were 112 g and 288 g respectively.

Use these results to show that the model may not be appropriate. [1]

- (e) Suggest a different value of a parameter of the model in the light of these results. [2]

2(a) The histogram shows information about the numbers of cars in five different price ranges, sold in one year at a car showroom.



It is given that 66 cars in the price range £10 000 to £20 000 were sold.

Find the number of cars sold in the price range £50 000 to £90 000.

[1]

(b) State the units of the frequency density.

[1]

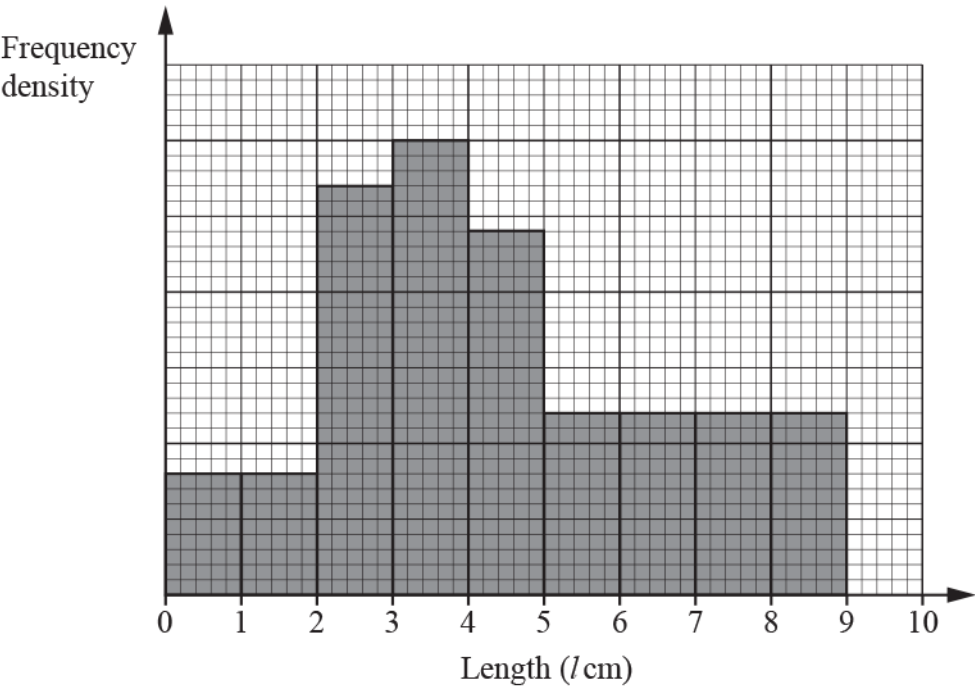
(c) Suggest one change that the management could make to the diagram so that it would provide more information.

[1]

(d) Estimate the number of cars sold in the price range £50 000 to £60 000.

[1]

3(a) The histogram shows information about the lengths, l centimetres, of a sample of worms of a certain species.

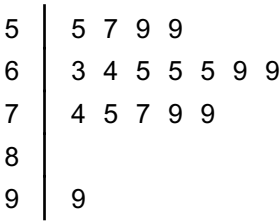


The number of worms in the sample with lengths in the class $3 \leq l < 4$ is 30.

Find an estimate of the number of worms in the sample with lengths in the range $4.5 \leq l < 5.5$. [3]

(b) Find the number of worms in the sample with lengths in the class $0 \leq l < 2$. [2]

4(a) The stem-and-leaf diagram shows the heights, in centimetres, of 17 plants, measured correct to the nearest centimetre.



Key: 5 | 6 means 56

Find the median and inter-quartile range of these heights. [3]

(b) Calculate the mean and standard deviation of these heights. [2]

(c) State one advantage of using the median rather than the mean as a measure of average for these heights. [1]

5(a) Joseph drew a histogram to show information about one Local Authority. He used data from the “Age structure by LA 2011” tab in the large data set. The table shows an extract from the data that he used.

Age group	0 to 4
Frequency	2143

Joseph used a scale of 1 cm = 1000 units on the frequency density axis.

Calculate the height of the histogram block for the 0 to 4 class. [2]

(b) Magdalene wishes to draw a statistical diagram to illustrate some of the data from the “Method of travel by LA 2011” tab in the large data set.

State why she cannot draw a histogram. [1]

6(a) The table shows information about the number of days absent last year by students in class 2A at a certain school.

Number of days absent	0	1	2 to 4	5 to 10	11 to 20	21 to 30	More than 30
Number of students	7	12	9	1	0	1	0

Calculate an estimate of the mean for these data. [2]

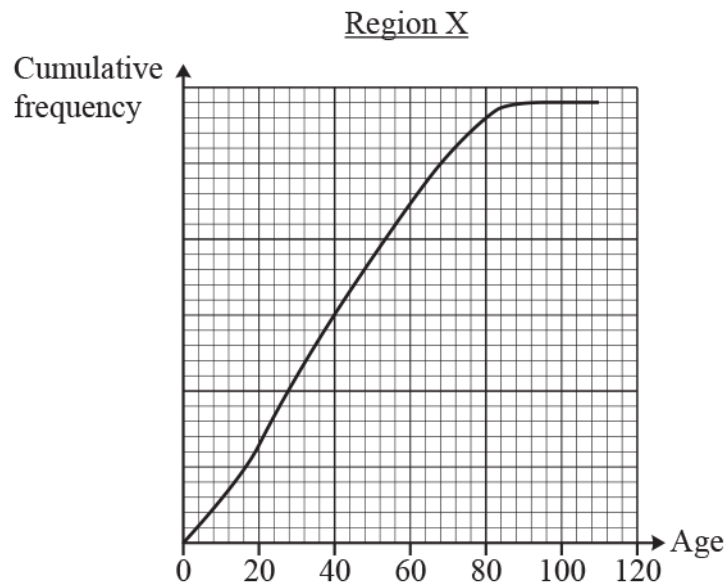
(b) Find the median of these data. [1]

(c) The headteacher is writing a report on the numbers of absences at her school. She wishes to include a figure for the average number of absences in class 2A. A governor suggests that she should quote the mean. The class teacher suggests that she should quote the median, because it is lower than the mean.

Give another reason for using the median rather than the mean for the average number of absences in class 2A. [1]

7 Paul drew a cumulative frequency graph showing information about the numbers of people in various age-groups

in a certain region X. He forgot to include the scale on the cumulative frequency axis, as shown below.

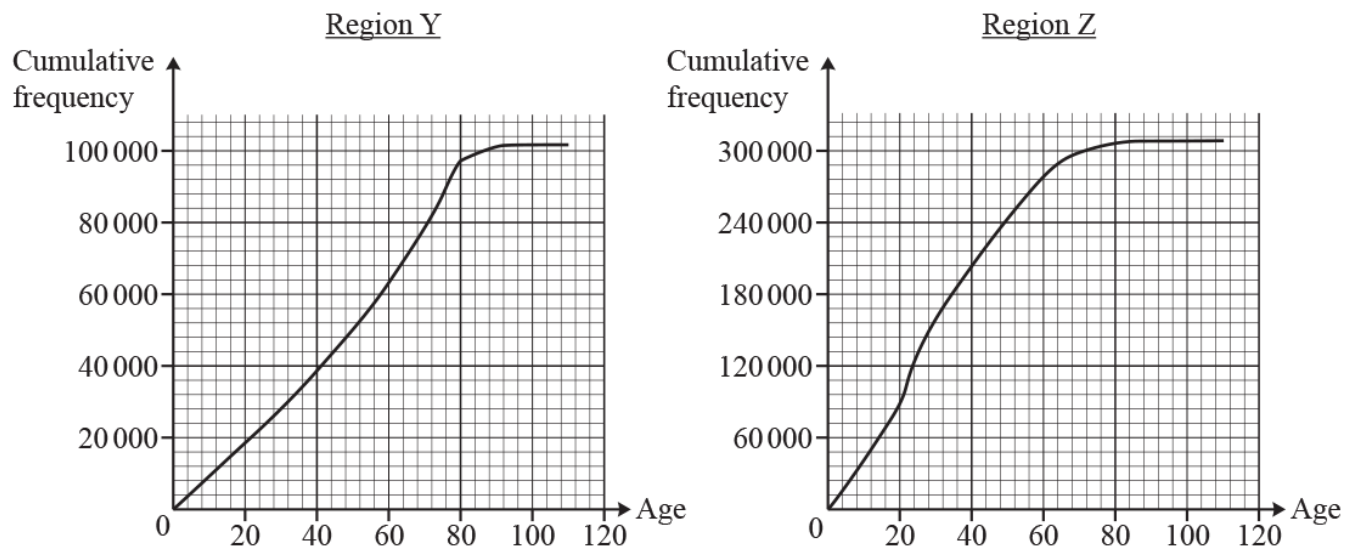


(a) Find an estimate of the median age of the population of region X. [1]

(b) Find an estimate of the proportion of people aged over 60 in region X. [2]

Sonika drew similar cumulative graphs for another two regions, Y and Z, but she included the scales on the

cumulative frequency axes, as shown below.



- (c) Find an age group, of width 20 years, in which region Z has approximately 3 times as many people as region Y. [1]

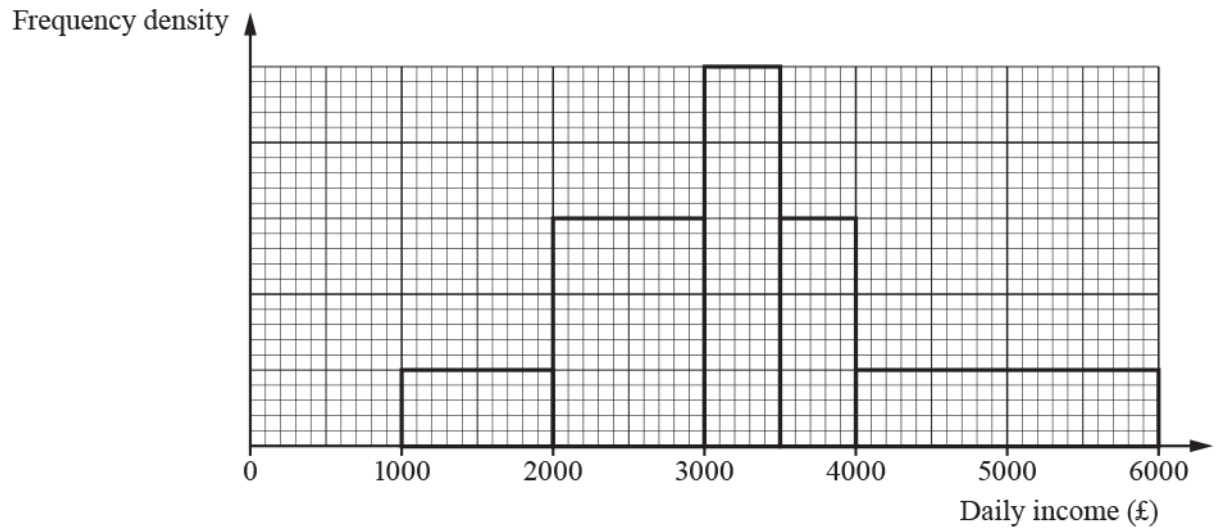
- (d) State one advantage and one disadvantage of using Sonika's two diagrams to compare the populations in Regions Y and Z. [2]

- (e) Without calculation state, with a reason, which of regions Y or Z has the greater proportion of people aged under 40. [1]

The marks of 24 students in a test had mean m and standard deviation $\sqrt{6}$. Two new students took the same test. Their marks were $m - 4$ and $m + 4$.

Show that the standard deviation of the marks of all 26 students is 2.60, correct to 3 significant figures. [3]

- 9 The finance department of a retail firm recorded the daily income each day for 300 days. The results are summarised in the histogram.



- (a) Find the number of days on which the daily income was between £4000 and £6000. [3]

- (b) Calculate an estimate of the number of days on which the daily income was between £2700 and £3600. [3]

(c) Use the midpoints of the classes to show that an estimate of the mean daily income is £3275.

[2]

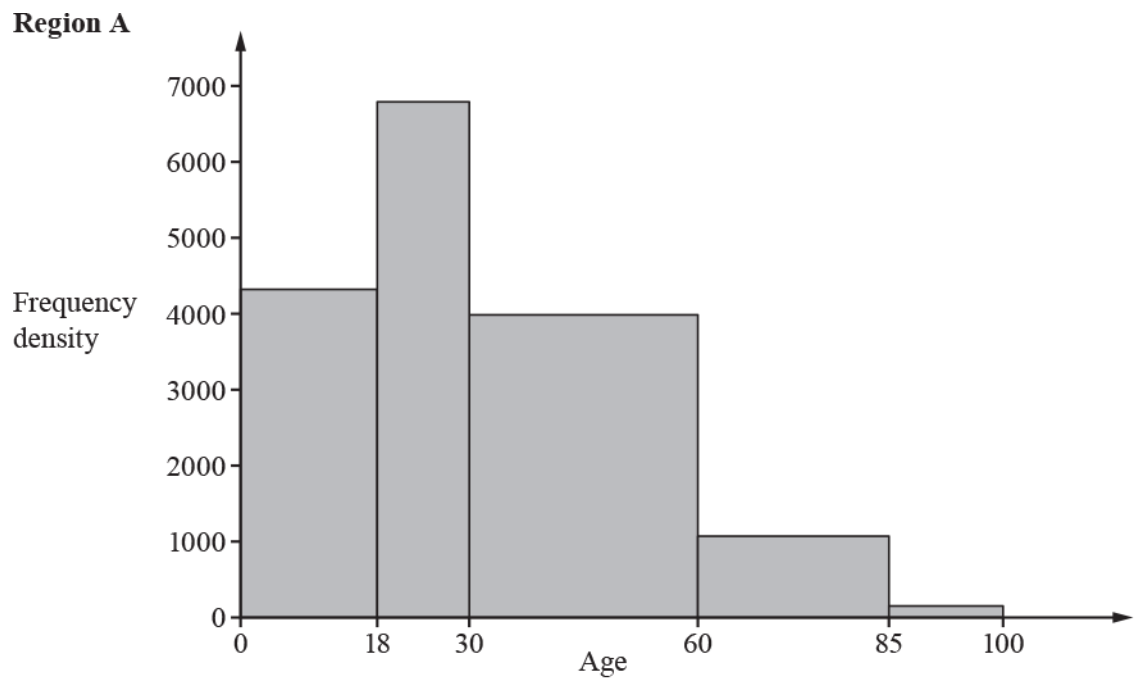
An estimate of the standard deviation of the daily income is £1060. The finance department uses the distribution $N(3275, 1060^2)$ to model the daily income, in pounds.

(d) Calculate the number of days on which, according to this model, the daily income would be between £4000 and £6000.

[2]

(e) It is given that approximately 95% of values of the distribution $N(\mu, \sigma^2)$ lie within the range $\mu, \pm 2\sigma$. [2]
Without further calculation, use this fact to comment briefly on whether the proposed model is a good fit to the data illustrated in the histogram.

10 John used data from the 2011 UK census to produce the following histogram for region A.



In the Census report, the age classes were given as follows.

0	5	8	10		16	18	20	25	30	45	60	65	75	85	90
to	to	to	to	15	to	to	to	to	to	to	to	to	to	to	and
4	7	9	14		17	19	24	29	44	59	64	74	84	89	over

John combined classes to give the classes shown in the histogram.

- (a) (i)

Explain the reason for John’s choice of upper class boundary for the first class.

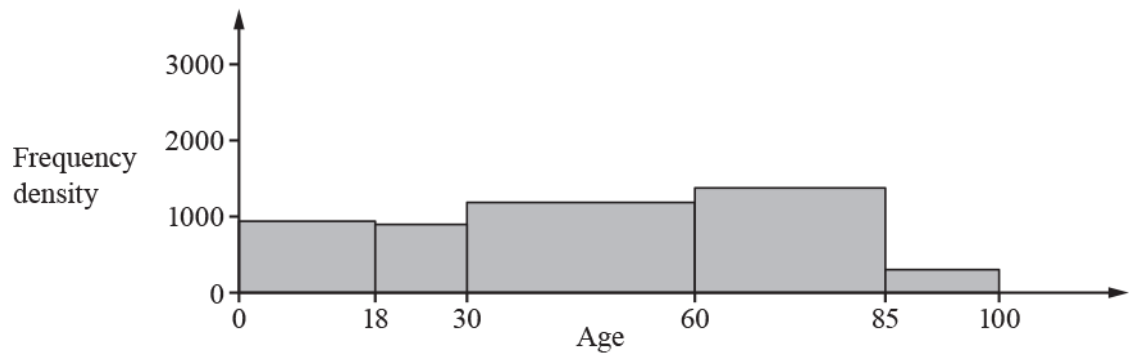
[1]
- (ii)

Suggest a reason for John’s choice of upper class boundary for the last class.

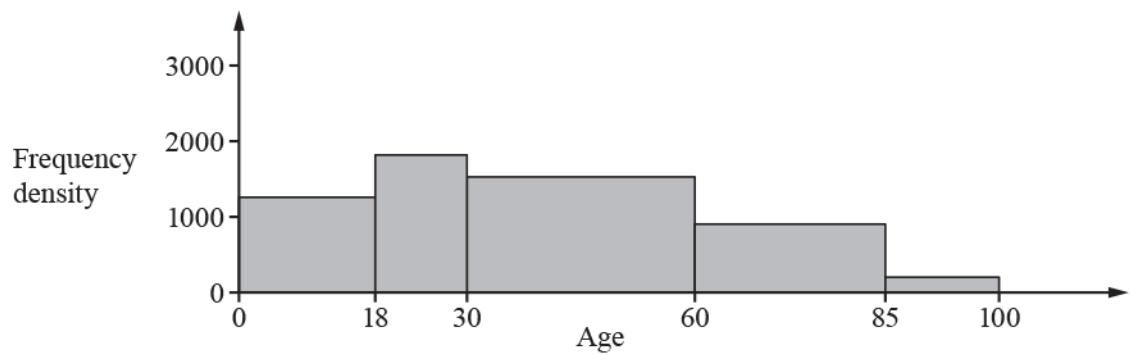
[1]

John also produced similar histograms for two other UK regions, B and C.

Region B



Region C



- (b) Which of the three regions had the largest proportion of people aged 85 and over? Without detailed calculations, explain your answer.

[3]

The mean ages, in years, of the populations in the three regions were 47.5, 39.5 and 31.5.

(c) For each of these means, state the region to which it corresponds. Justify your answers.

[3]

John made the following claim.

“The histograms show that a child living in region B in 2011 could expect to live longer than a child living in region A in 2011.”

(d) Is this claim justified? Give a reason for your answer.

[1]

- 11 Frances used the pre-release data set to produce the following table which shows information about the residents of Norwich in 2011.

Age	0 to 15	16 to 24	25 to 44	45 to 64	65 and over	Total
Number of residents	21 707	22 921	40 894	27 645	19 345	132 512

- (a) State the upper class boundary of the “25 to 44” class.

[1]

Frances used these data to calculate estimates of the mean and standard deviation of the ages of these residents. She assumed that the oldest resident was aged 105.

- (b) Calculate these estimates.

[3]

- (c) Use these estimates to discuss whether there may be any outliers.

[2]

- (d) Jacob suggested that more reliable estimates could be obtained by using the value 80 for the last class, instead of the midpoint. Explain, with a reason, whether you think this suggestion is a good one.

[1]

END OF QUESTION PAPER

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
1	a		(mean =) 201 (3 sf) (sd =) 60.7 (3 sf)	B1 (AO1.1) B1 (AO1.1) [2]	Allow 60.8
	b		0.364 (3 sf)	B1 (AO3.4) [1]	
	c		$P(X < 160) = 0.252(49)$ $x_1 = \Phi^{-1}(0.6 + '0.25249')$ $= 262.83$ (5 sf) ISW	B1 (AO3.4) M1 (AO1.1) A1 (AO1.1) [3]	soi, eg by $P(X > 160) = 0.748$ or 0.747 or by 0.147 or 0.148 T&I: correct answer scores B1M1A1, otherwise max B1 SC Answer 263 with correct working: B1M1A0 SC Answer 263 with inadequate working: B1 only
	d		112 and 288 are within 2 sd from mean (no working needed) $P(X < 112) = 0.0708$, which is > 0.025 or > 0.0013 or > 0	B1 (AO3.5a) [1]	or $\mu + 2\sigma = 320$ ($\mu + 3\sigma = 380$) which is $>$ than 288 or $P(112 < M < 288) = 0.858$ which is $<$ than 0.95 (or 0.99) or $p = 0.858$, but model suggests $p = 1$ NOT 0.858 alone B0
	e		Reduce σ $288 - 200 = 2\sigma$ or $288 - 200 = 3\sigma$ or $288 - 112 = 4\sigma$ or $288 - 112 = 6\sigma$ $\sigma = 44$ $\sigma = 29.3$ or about 30	B1 (AO3.5c) B1 (AO3.3) [2]	May be implied by value of σ Allow more precise correct methods Allow σ between 25 and 50. No working needed B1B1 or σ^2 between 625 and 2500 <u>Examiner's Comments</u> This question is best answered by noting that most (or almost all) the values of a normal distribution lie within 2 (or 3) standard deviations of the mean. However, many candidates who found these values did not explicitly compare them with the two values given in the question.
			Total	9	
2	a		$(40000 \times 0.002 =) 80$	B1 (AO1.1) [1]	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
	b		Frequency per £ or No. of cars per £	B1 (AO1.1) [1]	Allow cars / £ Allow as fraction NOT cars / price NOT cars / money Examiner's Comments This was an unusual type of question, and many candidates did not understand what was required. Common incorrect answers were $\frac{\text{Frequency}}{\text{Class width}}$, $\frac{\text{Cars}}{\text{Price}}$ or $\frac{\text{Number of cars}}{\text{Thousands of pounds}}$
	c		Show fds for four separate classes within £50000 - £90000	B1 (AO2.3) [1]	Make price intervals Show the smaller Include more bars median (or similar) Give a title Explain y-axis better or give units of f.d. Give the frequencies
	d		20	B1f (AO1.1) [1]	or (a) ÷ 4
			Total	4	
3	a		Freq 4-5 = 24 or freq 5-6 = 12 $\frac{1}{2}(\text{freq 4-5} + \text{freq 5-6})$ or $\frac{1}{2} \times 24 + \frac{1}{2} \times 12$ = 18	B1(AO3.1a) M1(AO1.1) A1(AO1.1) [3]	or freq 5-9 = 48 OR similar with frequency density or $\frac{1}{2}(\text{freq 4-5}) + \frac{1}{8}(\text{freq 5-9})$ oe Correct answer without working or unclear working: allow B1M1A1
	b		$30 \times 2 \times 1.6/6$ or $30 \times 2 \times 8/30$ oe = 16	M1(AO1.1) A1(AO1.1) [2]	or $15 \times 80/150$ oe Correct answer without working or unclear working: allow M1A1
			Total	5	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
4	a		65 Quartiles are 76 and 61 = 15	B1(AO1.1) M1(AO1.1) M1(AO1.1) [3]	Allow (75 to 77) and (59 to 63) Must come from 76 – 61 SC Misread 5 6 = 5.6: lose B1 only Examiner's Comments Most candidates found the median correctly. However, many candidates used a slightly incorrect method to find the quartiles. For example: $\frac{17}{4} = 4.25$, so the Lower Quartile is 60. Textbooks vary in the method for finding the quartiles, but in the case of these 17 heights, any correct method leads to the value 61cm. Perhaps the simplest way to arrive at this is to find the median of the lowest 8 heights.
	b		Mean = 69 sd = 10.5 (3 sf)	B1(AO1.1) B1(AO1.1) [2]	Allow 6.9 Allow 1.05 (3 sf) Examiner's Comments Most candidates found both values correctly, although some made arithmetical errors. Those who simply entered the values in the calculator and used the relevant functions were generally successful. Those who used the "long-hand" method sometimes made arithmetical errors. It is worth noting that there were only two marks for this question, which is a strong pointer to the fact that candidates are free to use their calculator functions, without showing working, as is the command word 'Calculate'.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
	c		Less (or not) affected by the outlier or anomaly 99 Mean (more) affected by the outlier of 99	B1(AO1.1) [1]	oe, but must mention 99 Allow "Median is less skewed by the 99" Ignore all else, eg "more accurate" <u>Examiner's Comments</u> Questions almost identical to this one have been asked (and commented upon in Examiners' reports) for many years, in legacy paper Statistics 1 (4732). Nevertheless, many candidates made the usual error of giving a correct generic answer, but not related to the context of the question data. Many wrote statements that began with "If there are any outliers..." without apparently having considered whether there <u>are</u> actually any outliers in the data in this question. Some other candidates gave inadequate reasons such as "The median is the middle value, and so is more representative". Some claimed that the median is "more accurate" for one of a number of possible reasons.
			Total	6	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
5	a		$\frac{2143}{5}$ or 428.6 or 429 Height = 0.429 (3 sf) (cm)	M1(AO1.1a) A1(AO1.1) [2]	M1 for $\frac{2143}{4}$ or $\frac{2143}{4.5}$ or 535.75 or 534 or 0.534 Allow without "cm" or $\frac{2143}{4.5}$ or 476 or 0.476 eg $\frac{1000}{428.6}$ M1A0 <u>Examiner's Comments</u> The vast majority of candidates used a class width of 4. The correct class width is 5, because "Age group 0 to 4" is shorthand for $0 \leq \text{age} < 5$. Some candidates did not use the class width but just divided the frequency by 1000. A few multiplied the frequency by the class width.
	b		Non-numerical data oe	B1(AO1.2) [1]	Indication of numerical data needed, eg Boundaries need to be between 2 numbers Cannot be measured. Qualitative data. No range of values NOT "No class boundaries", "No class widths", "Not g'ped data", "Need continuous data" "No range" Ignore all else <u>Examiner's Comments</u> Any response that stated or implied that "Method of travel" is non-numerical was accepted. However such responses were few and far between. Many and various incorrect or inadequate responses were seen, including the following. "There are no class widths", "There are no ranges", "The data is not grouped", "The data is discrete", "The data is not continuous", "There is too much data for a histogram", "The frequency density would be difficult to display", "You cannot calculate the frequency density", "There is no gap between the bars" and "There is insufficient information to draw a histogram".
			Total	3	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
6	a		$\frac{\Sigma fx}{\Sigma f}$ $= 2.4$ $(\text{= } \frac{72}{30})$	M1(AO1.1a) A1(AO1.1) [2]	attempted. May be implied by working seen or eg ≥ 2 terms in numerator correct Dep $\div \Sigma f$ (NOT = 7) Examiner's Comments Most candidates used the correct method. Some used incorrect mid points, or even end points. Some thought that $7 \times 0 = 1$. A large minority found $\frac{\Sigma f}{7}$.
	b		1	B1(AO1.2) [1]	Ignore method except 0, 0, 1, 1, 7, 9, 12 or 7, 12, 9, 1, 0, 1, 0 B0 Examiner's Comments Most candidates answered correctly. Some found the 15th value, rather than the mean of the 15th and 16th values, but this error was condoned. A few found the median of the frequencies. This unfortunately gives the correct response, but it did not gain the mark.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
	c		The median is less influenced than the mean by the one student in the 21 - 30 class	B1(AO3.4) [1]	or equivalent statement. Must clearly refer to the actual outlier in 21 - 30 class Not "It's the same as the mode". Ignore "skew". <u>Examiner's Comments</u> Surprisingly this question was less well answered than similar questions on Paper S1 in the past. Many candidates gave spurious reasons such as that the median is an integer whereas the mean is not, or the median is the same as the mode, or the median is not an estimate, or the median is the middle value, or the median is easier to calculate. Many others quoted from the text book that the median is less affected by outliers, but did not consider whether there are actually any outliers in this case. To gain the mark the response had to be in the context of the question, stating that there is an actual outlier, and to specify which candidate this is.
			Total	4	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
7		a	38 to 39	B1 (AO 3.1a) [1]	
		b	eg $\frac{1.25}{5.75}$ or $\frac{7}{29}$ = 0.2 to 0.24	M1 (AO 3.1a) A1 (AO 1.1) [2]	Use heights, any units, eg cm or squares
		c	eg 40 to 60	B1 (AO 1.1) [1]	Any correct range
		d	Can easily compare proportions or age profile Cannot easily compare numbers in age groups	E1 (AO 2.2b) E1 (AO 2.2b) [2]	
		e	Z. Graph steeper for below 40	E1 (AO 2.2b) [1]	
			Total	7	
8			$\frac{\sum(x-m)^2}{24} = (\sqrt{6})^2$ $\sum(x-m)^2 = 144$ Mean of all 26 is m $S^2(26) = \frac{144+2 \times 4^2}{26}$ $= \frac{176}{26} \text{ or } \frac{88}{13}$ $S = \sqrt{\frac{88}{13}} \quad (= 2.60 \text{ (3 sf) AG})$	M1 (AO 3.1a) M1 (AO 1.1a) A1 (AO 1.1) [3]	$\frac{\sum x^2}{24} - m^2 = (\sqrt{6})^2$ $\sum x^2 = 144 + 24m^2$ $S^2(26) = \frac{144+24m^2+(m-4)^2+(m+4)^2}{26} - m^2$ $= \frac{176}{26} \text{ or } \frac{88}{13}$ Must see $\sqrt{\frac{88}{13}}$ or equivalent and ans 2.60
			Total	3	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance	
9		a	Total area = 500 small squares $\frac{100}{500} \times 300$ = 60 (days)	B1 (AO1.2) M1 (AO1.1a) A1 (AO1.1) [3]	or 20 cm ² or other units or $\frac{4}{20} \times 300$ or equivalent	May be implied
		b	$\frac{3 \times 15 + 5 \times 25 + 1 \times 15}{500} \times 300$ = 111	M1 (AO2.1) M1 (AO1.1) A1 (AO1.1) [3]	M1 for denom & one term in num M1 M1 for correct $\times$ 300	oe in other units
		c	Frequencies: 30, 90, 75, 45, 60 $\frac{\Sigma fx}{300} = \frac{982500}{300}$ (= 3275 AG)	M1 (AO3.1a) A1 (AO1.1) [2]	Allow multiples of these correctly obtain 3275	
		d	$P(4000 < x < 6000) \times 300$ = $0.2419 \times 300 = 72.57$ so 73 days	M1 (AO3.4) A1 (AO1.1) [2]	Attempted, using $N(3275, 1060^2)$ BC accept truncation to 72 days	
		e	$3270 + 2 \times 1060 = 5395$ In histogram, well over 2.5% of values are above 5395, so model not a good fit	B1 (AO3.4) E1 (AO2.2b) [2]		
			Total	12		

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance	
10		a	(i) "0 to 17" means $0 \leq \text{age} < 18$	E1(AO 1.2) [1]	Allow "17" means up to 17 yrs, 364 days	or any correct
		a	(ii) Original class had no UCB but for histogram an UCB was needed. Few people live > 100	E1(AO 1.1) [1]	or other sensible reason	
		b	B has more aged 85 - 100 than A, which has larger total B & C have similar totals, and there are more aged 80 - 100 in B B	E1(AO3. 1a) E1(AO2. 4) B1(AO2. 2a) [3]	dep on E1E1 earned	
		c	A: 31.5; B: 47.5; C: 39.5 A has greatest proportion of area towards left B has greatest proportion of area towards right	E1(AO 2.2a) E1(AO 3.1a) E1(AO 2.4) [3]	or similar correct explanation	
		d	Not justified B has higher mean than A But could be caused by older people moving away from A, or moving into B	B1 [1]	or B has higher proportion of older than A or eg people move to B to retire	or other sensible
			Total	9		

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
11		a	45	B1(AO3.1b) [1]	Allow 44 years 364 days or similar
		b	$\bar{x} = 39.6$ (3 sf) $s = 24.3$ (3 sf)	B1(AO1.1) M1(AO1.1) A1(AO1.1) [3]	(UCB = 106 because oldest is 105) (If seen) sub in correct formula using any x 's within classes Allow 24.2 (3 sf) If use 105 as UCB: mean = 39.5 (3 sf) B0 sd = 24.2 (3 sf) M1A1
		c	$39.6 + 2 \times 24.3 = 88.2$ $39.6 - 2 \times 24.3 = -9$ Hence <u>may be</u> outliers at top, but not at bottom	M1(AO1.1a) A1f(AO2.2b) [2]	Allow just $39.6 + 2 \times 24.3 = 88.2$ for M1 A1 for both limits and full conclusion ft their $\bar{x}$ and s
		d	No, most in 65+ class will be nearer the lower end oe	E1(AO3.2b) [1]	or imply class is weighted towards left
			Total	7	